SCIENCE · YEAR 10 · KS4 BIOLOGY

Photosynthesis

Learning Booklet

Name: __________________ Class: ________ Date: ________

What you are learning

- Say what photosynthesis is and where it happens.
- Write the word equation (and the symbol equation).
- Name what goes in (reactants) and what comes out (products).
- Describe how light, carbon dioxide and temperature change the rate.

Common mistakes (read first)

- Mixing up the gases. A plant takes IN carbon dioxide and gives OUT oxygen — not the other way round.
- Calling light a 'raw material'. The raw materials are water and carbon dioxide; light is the energy source.
- Saying it happens 'in the leaf'. Be exact: it happens in the chloroplast inside the leaf cell.
- Forgetting plants respire too. Plants photosynthesise AND respire (respiration never stops, day or night).
- Writing the equation back to front. Reactants (CO_2 + water) are on the LEFT, products (glucose + oxygen) on the RIGHT.

Key vocabulary

Photosynthesis	how a plant makes its own food (glucose) using light energy
Chlorophyll	the green pigment that absorbs (traps) light energy
Chloroplast	the part of a plant cell where photosynthesis happens
Glucose	the sugar the plant makes; used for energy and growth
Reactants	what you start with: carbon dioxide and water
Products	what is made: glucose and oxygen
Stomata	tiny holes under the leaf that let gases in and out
Limiting factor	the thing in shortest supply that slows the rate down

How photosynthesis works

Plants cannot move to find food, so they make their own. They do this using light energy from the Sun.

Light energy is trapped by chlorophyll, the green pigment inside chloroplasts. The energy is used to join carbon dioxide (from the air) and water (from the soil) together.

This makes glucose, a sugar the plant uses for energy and growth. Oxygen is made as well and is released into the air.

Word & symbol equation

carbon dioxide + water □(light + chlorophyll)□ glucose + oxygen

$6CO_2 + 6H_2O$ □ $C_6H_{12}O_6 + 6O_2$

Diagram 1

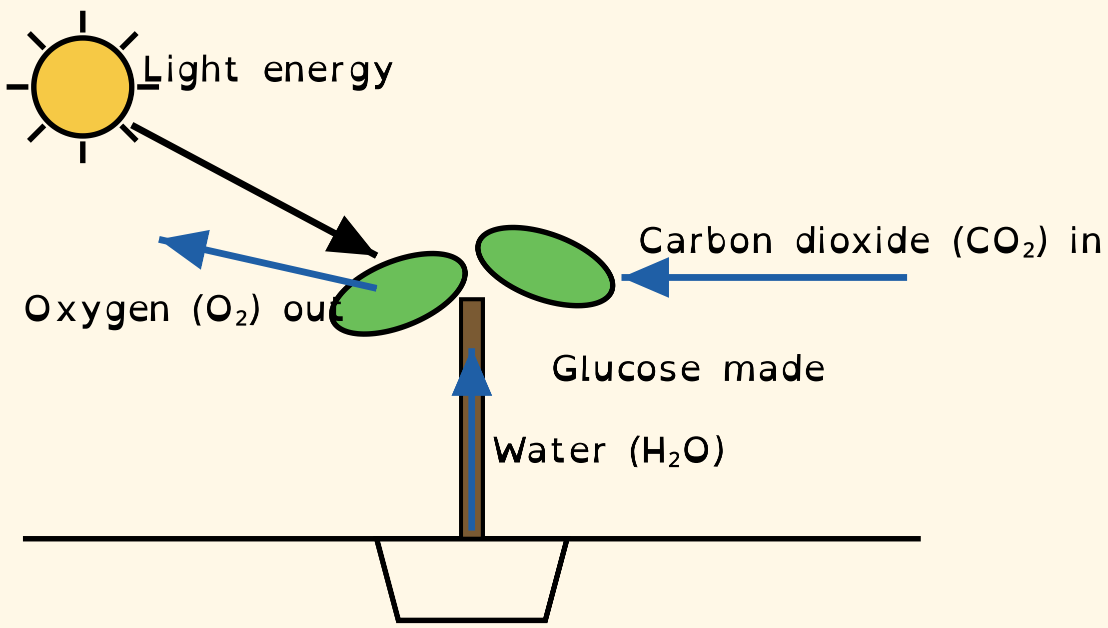

Diagram 1: What goes into and comes out of a plant

□ Remember (content)

- IN: carbon dioxide + water. OUT: glucose + oxygen.
- Where: chloroplasts in the leaf. Energy from: light, trapped by chlorophyll.

- Plants photosynthesise AND respire — respiration never stops.

Diagram 2

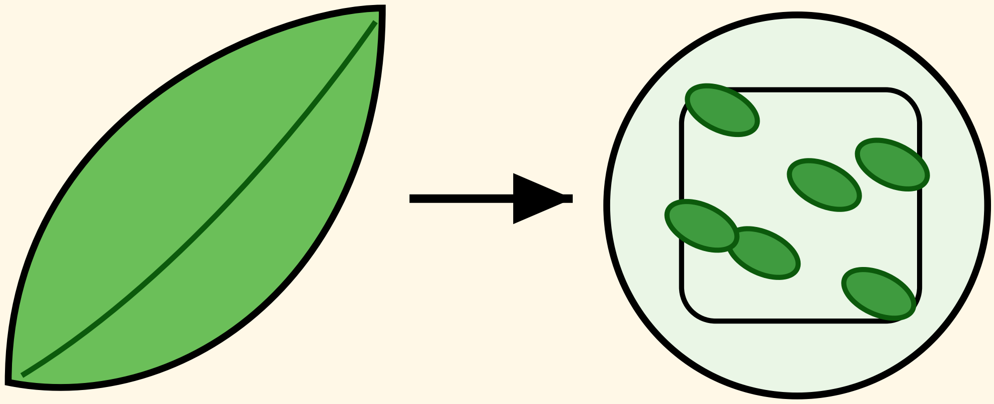

Diagram 2: Light is trapped by chlorophyll in the chloroplast

What the glucose is used for

- Released in respiration to give the plant energy.
- Used to make new cells for growth.
- Stored as starch for later.
- Turned into cellulose to build strong cell walls.

What changes the rate? (limiting factors)

- Light intensity — more light means a faster rate, until something else runs short.
- Carbon dioxide — more CO_2 means a faster rate, up to a point.
- Temperature — warmth speeds it up, but too hot damages the enzymes and it slows down.

Diagram 3

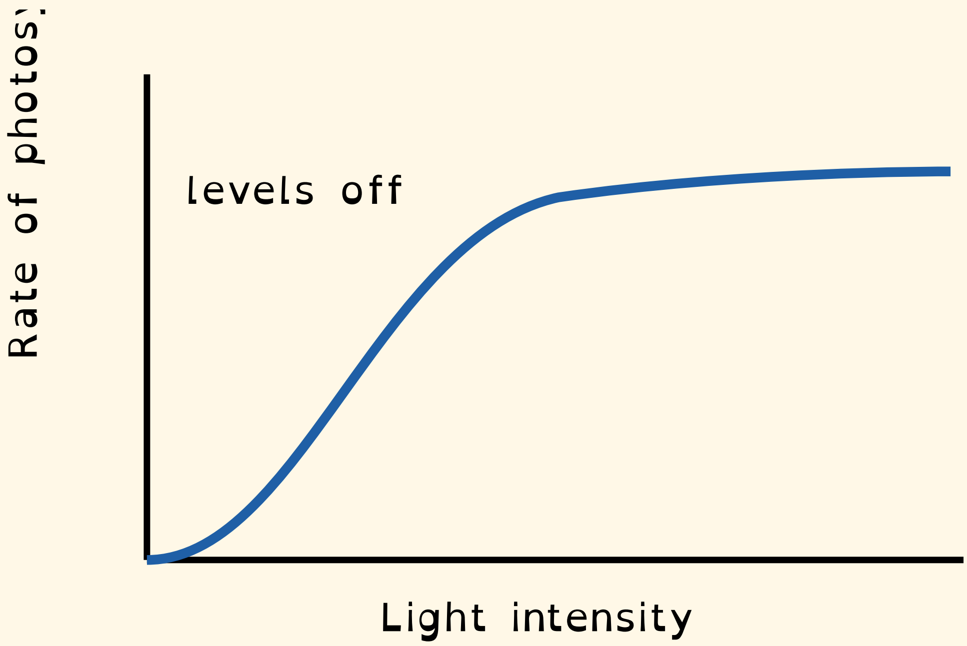

Diagram 3: Rate of photosynthesis vs light intensity (it levels off)

▯ How to answer an 'explain' question

- 1 Say what happens (the fact).
- 2 Say WHY using a key word (light / chlorophyll / chloroplast / rate).
- 3 Link it back to the question (so… / this means…).
- Example: 'No light, so chlorophyll cannot trap energy, so no glucose is made.'

Questions — they get harder as you go

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Q1 **[1]** Recall ●○○○○

Which gas does a plant take in for photosynthesis?

Q2 **[1]** Recall ●○○○○

Which gas does a plant release?

Q3 **[1]** Recall ●●○○○

Name the green pigment that traps light.

Q4 **[2]** Recall ●●○○○

Write the word equation for photosynthesis.

Q5 **[2]** Apply ●●●○○

Give two things a plant uses glucose for.

Q6 **[3]** Explain ●●●●○

A plant is left in a dark cupboard for a week. Explain why it cannot photosynthesise.

Q7 **[4]** Analyse (stretch) ●●●●●

A student shines a brighter light on some pondweed and counts more oxygen bubbles. Explain why the rate increases, then explain what could stop it increasing any further.

Self-reflection

How confident do you feel with photosynthesis now? (tick one)

☐ Not yet ☐ Getting there ☐ Confident ☐ Really confident

One key word I want to remember:

____

____

The question I found hardest was ____ because:

____

____

One thing I will revise before the next lesson:

____

____

Teacher assessment

Success criteria (tick if met) — total: 14 marks

- ☐ Names the gases correctly (in: CO_2, out: O_2)
- ☐ Writes the word equation correctly
- ☐ States where photosynthesis happens (chloroplast)
- ☐ Explains using key words (light / chlorophyll / rate)
- ☐ Explains a limiting factor (stretch)

Red — needs re-teaching | Amber — nearly there | Green — secure

What went well (WWW):

___

Even better if (EBI):

___

Next step / target:

___

Teacher answer key

1. Q1: Carbon dioxide.
2. Q2: Oxygen.
3. Q3: Chlorophyll.
4. Q4: carbon dioxide + water → glucose + oxygen (light + chlorophyll above the arrow).
5. Q5: Any two of: respiration/energy, growth, stored as starch, made into cellulose.
6. Q6: No light energy, so chlorophyll cannot trap energy; without light the reaction cannot run, so no glucose is made.
7. Q7: More light = more energy = faster rate, so more oxygen made. It stops increasing when another factor becomes limiting (carbon dioxide, temperature) or light is no longer the limiting factor.